CO_2 Capture Model from the Science and Technology Center using an Aqueous Calcium Hydroxide Slurry Sorbent

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Introduction

Climate change is mainly caused by the rising atmospheric CO_2 levels, which have increased and reached their highest concentration in over two million years, according to the Intergovernmental Panel on Climate Change (IPCC, 2021). This increase is directly related to the combustion of fossil fuels, which is the main source of global greenhouse gas emissions. Elevated CO_2 levels have become a serious global issue as they contribute to more frequent extreme weather events, changes in agricultural activity, and create long-term conditions that directly affect human health and safety. These impacts are not limited to specific places, they affect communities everywhere. Since CO_2 remains in the atmosphere for extended periods of time, every additional emission adds to these long-lasting implications. This makes reducing CO_2 emissions very important for protecting both current and future populations.

Universities have an important role in this problem, as well as in possible solutions. Their campuses include buildings that are energy-intensive and require substantial heating, cooling, and ventilation, especially in facilities that contain laboratories and research equipment, where environmental control is necessary. These utilities often rely on systems powered by fossil fuels such as natural gas. Tufts University, through its commitments to the Climate Leadership Network and the Association for the Advancement of Sustainability in Higher Education (AASHE), has committed to reduce its greenhouse gas emissions and work toward long-term carbon neutrality by 2050 (AASHE, 2024; Second Nature, 2024). To reach these goals, it is needed to identify major places that are sources of emissions within campus and explore strategies that can reduce CO_2 emissions without affecting building operations.

This project focuses on the Science and Technology Center (SciTech) at Tufts University, located at 4 Colby Street. According to the ChBE 39 project description, SciTech runs on a natural gas boiler that is the primary source of the building's direct CO_2 emissions, and replacing this system with a renewable option is not currently feasible due to budgetary and operational constraints. This challenge creates an opportunity to investigate alternative solutions, such as CO_2 capture using a reactive sorbent, a particularly relevant for natural-gas-based systems with flue gas with typical CO_2 concentrations around 8–12%; these concentrations are well suited for aqueous or slurry-based scrubbing methods (EPA, 2023). Capturing CO_2 before it is released into the atmosphere would allow Tufts University to significantly reduce SciTech's emissions while keeping the existing heating system in place.

To address the challenge of capturing for emitted CO_2, this project will focus on calcium hydroxide ($Ca(OH)_2$) in an aqueous slurry as the sorbent. Aqueous chemical sorbents have recently demonstrated both technical viability and scalability in industrial applications, which presents new opportunities for CO_2 emission reduction (NIST, 2023).

Calcium hydroxide is a strong base that reacts with CO_2 through acid-base chemistry to form calcium carbonate and water. This fundamental reaction proceeds as follows:

$$Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O \qquad [1]$$

This is a thermodynamically favorable reaction and it can operate at ambient temperatures and atmospheric pressures, making it suitable for retrofit applications with limited chances of major infrastructural changes and design. Aqueous sorbents like this one have been on the rise in the field of carbon capture because a liquid phase allows for fast diffusion of CO_2 into the bulk sorbent, reducing the time required for significant absorption compared to traditional solid sorbents. Additionally, calcium hydroxide is relatively inexpensive and widely available, making it a desired sorbent subject (NETL, 2015).

The primary limitation of aqueous $Ca(OH)_2$ systems is the limits imposed from equilibrium capture. At higher temperatures and pressure, the reaction approaches equilibrium and the driving force for absorption is slowed down. This means that the overall absorption efficiency is typically around 11.83 to 19% in these systems (Han et al., 2011). Another factor reducing effectiveness of the system is the formation of the precipitate product calcium carbonate; it can reduce the effective sorbent concentration over time and potentially cause equipment fouling. These constraints mean that we need a process design capable of balancing sorbent capacity utilization, operating temperature, and residence time within the capture reactor. The objective of this project will be to model the varying relationships between said equilibrium constraints and transport kinetics to quantify the performance of a theoretical reactor and to develop a design that maximizes CO_2 removal while maintaining operational feasibility.

The scope of this project centers on modeling and designing a capture mechanism. Our goal is to develop a differential-equation-based model that represents the key thermodynamic and kinetic processes involved in CO_2 absorption and carbonate formation in the slurry. With this model, we plan to analyze how changes in system parameters; such as sorbent concentration, particle size, mixing intensity, or residence time, affect overall CO_2 removal efficiency. We will not be focusing on topics such as handling or storing the calcium carbonate byproducts or the

physical installation of a capture unit in SciTech, as these aspects fall outside the scope of our project.

Overall, this project focuses on creating and modeling a CO_2 capture system for the SciTech building to help reduce the emissions it produces. With this, we hope to contribute to ongoing changes at Tufts to lower carbon emissions and raise awareness of the issue, which affects all of us in the long term.

Reactor Model Definition

SciTech Emissions Composition Breakdown

According to the North American Energy Standards Board, natural gas is made up of 94.9 mol % methane as its main component. We will thus assume that the natural gas consumption from the SciTech natural gas boiler consisted of 95% methane, and will neglect the other gaseous components in this mixture. The next assumption for our model will be that there is a complete combustion reaction from CH_4 to CO_2, obeying the following combustion reaction equation:

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O\ , \quad \Delta H_{rxn} = -\ 890.8\ \text{kJ/mol} \quad [2]$$

This should be a safe assumption, given that properly tuned boilers should convert 99.9 percent of natural gas into CO_2 during the combustion process (US EPA, 1.4). Given these simplifications, and given a heat of combustion of -890 kJ/mol, it is possible to obtain an estimated molar flow rate of CO_2 from the SciTech natural gas boiler by analyzing energy consumption data. Averaging yearly energy consumption numbers from 4 Colby Street for the fiscal year 2024, it was translated to an average of 1273.37 mol CO_2 /hr, or 56.04 kg CO_2/hr (Tufts University Energy Public Data). Given this value, we can now consider this to be our inlet gas flow rate for the remainder of our model.

Considering typical values for natural-gas boiler exhausts, the **inlet CO_2 mole fraction** in flue gas for a reactor that could be used with the SciTech infrastructure is about **10%** (Song et al., 2004). This would yield a **total molar flow rate of 12733.7 mol/hr**. This will also be used in our calculations and will be relevant when establishing the actual inlet CO_2 fraction, which for the purpose of this project will be required to be below 2 w% at the outlet.

Governing Equation and Liquid Model for Sorbent

The nature of the aqueous $Ca(OH)_2$ slurry is a tricky one. Although it is inherently in aqueous solution, it does not strictly abide by a liquid model. The slurry absorbs CO_2 through a carbonation reaction that forms solid calcium carbonate $CaCO_3$, which is described as a white, powdery limestone, effectively capturing the CO_2 and removing it from the air. It is best represented by the following reaction:

$$Ca(OH)_{2\,(s)} + CO_{2\,(g)} \rightarrow CaCO_{3\,(s)} + H_2O_{(l)}, \quad \Delta H_{rxn} = -109 \text{ kJ/mol} \quad [3]$$

Thus, there are both solid and liquid components to be considered for this reaction model. However, we will operate under the simplifying assumption that this reaction is best categorized as a sorbent model of liquid falling drops. Any calculations that follow will be performed under the assumption that this model is a reasonable fit to the reality of CO_2 absorption from our aqueous slurry.

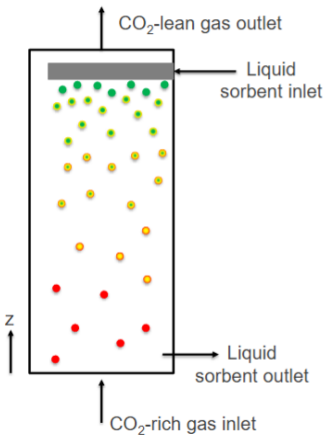


Figure 1. Cartoon Representation of Liquid Falling Drops Sorbent Model (Leverick, 2025)

Given this, some system parameters must be defined to carry out a realistic evaluation of the model. The most important values to solve for are the concentration of CO_2 in the gas phase at any given time t and position z along the reactor, as well as the concentration of unreacted sorbent in the liquid phase at any given time and position, and radius of spherical sorbent droplet r. These values will be referred to as $C_g(z,t)$ and $C_s(z,t,r)$, respectively. Radius of sorbent droplets r becomes a relevant parameter when considering the variance in radial compositions as the reaction is carried out; we will consider all drops at a given z and t to have the same radial distribution. At the inlet, $C_s(z,t,r)$ can be considered to be fully unreacted, C_s^o. Once fully reacted, we can treat that concentration to be 0 for all r.

From this theoretical foundation follow the governing partial differential equations (PDEs) that we will use to obtain these values. To describe the CO_2 concentration in the gas phase:

$$\varepsilon_g \frac{\partial C_g(z,t)}{\partial t} + u_g \frac{\partial C_g(z,t)}{\partial z} = -\frac{3(1-\varepsilon_g)}{R_p} k_{rxn} C_g(z,t) C_s(z,t,R_p) \quad [4]$$

where ε_g gives the fraction of the reactor volume that is gas (and the remainder is the sorbent drops); u_g is the superficial gas velocity, given by the gas volumetric flow rate $\hat{v}$ divided by the reactor area ($u_g = \frac{\hat{V}}{A}$); R_p is the radius of sorbent droplets; and k_{rxn} is the rate constant for the CO_2 and $Ca(OH)_2$ reaction.

Additionally, we get one more PDE that describes the behavior of this model:

$$\frac{\partial C_s(z,t,r)}{\partial t} + u_l \frac{\partial C_s(z,t,r)}{\partial z} = D\left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial C_s(z,t,r)}{\partial r}\right)\right] \quad [5]$$

Two useful boundary conditions can be applied to this equation:

$$\left.\frac{\partial C_s(z,t,r)}{\partial r}\right|_{r=0} = 0 \quad [6]$$

$$\left. -D\frac{\partial C_s(z,t,r)}{\partial r}\right|_{r=R_p} = k_{rxn} C_g(z,t) C_s(z,t,R_p) \quad [7]$$

where D is the diffusivity constant for the reacted sorbent into the sorbent droplet, and u_l is the droplet velocity.

A preliminary assumption might be that, under a liquid droplet model, time dependence for the modeled equations becomes negligible when considering the fast speeds of the reaction within a reasonably-sized reactor. This would further simplify this system of PDEs for a more manageable derivation approach. Another design choice that would allow for steady conditions would be to consider the case of a constant, fresh supply of calcium hydroxide slurry. Recycled slurry would affect the reacted sorbent concentration and make it increase over time; thus, a constant fresh supply would mean that the initial concentration of reacted sorbent will not be a time-dependent variable, and we can once again neglect time dependence for our system.

Given this assumption, equations [4] and [5] reduce to these simplified equations:

$$u_g \frac{\partial C_g(z)}{\partial z} = -\frac{3(1-\varepsilon_g)}{R_p} k_{rxn} C_g(z) C_s(z, R_p) \qquad [8]$$

$$u_l \frac{\partial C_s(z,r)}{\partial z} = D\left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(\frac{\partial C_s(z,r)}{\partial r}\right)\right] \qquad [9]$$

effectively dropping any time-dependent partial derivatives and the time-dependence of both C_g and C_s. To obtain a useful form of these derivatives for our models, we will use a finite difference approximation method to derive the concentration gradients across the reactor. Using the approaches learned in class, equations [8] and [9] respectively become:

$$C_{g,i+1} = C_{g,i} - \alpha(C_{g,i} C_{s,i}) \text{ , where } \alpha = \Delta z \frac{3(1-\varepsilon_g)}{R_p u_g} k_{rxn} \quad [10]$$

$$C_{i+1,j} = C_{i,j} + \frac{D\,\Delta z}{u_l\, r_j^2\, \Delta r^2}\left(C_{i,j+1} - 2C_{i,j} + C_{i,j-1}\right) \qquad [11]$$

When implementing these PDEs in our Spyder Python interface, we were provided with code from the instructor as a starting point for the analysis. In the following section, we will discuss the selection of critical parameter values based on relevant literature research and their subsequent implementation in the coding software to analyze and visualize the concentration gradients.

Critical Parameter Selection

To obtain a functional model, the following parameters must be defined:

A. Gas flow and volumetric flow rate (CO_2)
B. Reactor size (Diameter, Length and Volume)
C. Sorbent particle size, (R_p = radius of droplet)
D. Operating Temperature
E. Gas fraction in the reactor (ε_g)
F. Diffusivity of CO_2 in the sorbent phase D
G. Reaction rate constant k_{rxn}

These parameters were defined according to literature values and based off of previous calculations relevant to the SciTech natural gas boiler emissions. A summary of these parameters is reported in the Table below, and detailed explanations for each can be found in the Appendix.

Table I. Parameters used for our sorbent model

Parameter	**Value**
Gas molar flow rate	5.54 mol/s.
Gas volumetric flow rate	0.0865 m3/s.
Contact time	4.9 seconds
Reactor diameter	0.60 meters
Reactor height (L)	3.7 meters
Reactor Volume	1.06 m
Sorbent particle radius	10 μm
Temperature	298 K
Diffusivity of CO_2 in droplet	$2.0\times10-9\ m^2/s$
Reaction Rate Constant	$3\times10^{-8}\ m^5/(mol^2 \cdot s)$
Gas velocity	0.3 m/s
Liquid velocity	0.3 m/s
Inlet CO_2 concentration	4.1 mol/m^3

With these values now initially defined, we can begin to analyze our proposed model using the governing equations. An important note is that we are using the values in Table 1. as a starting point to have working code, but these will evolve through multiple iterations and as the model grows in complexity. Final, optimized parameters will be reported in the results section.

Modeling the $Ca(OH)_2$ Reactor with Code

Using the code provided by Professor Leverick formulated for a steady state system, our main workflow was to modify this code and add relevant complexities to test the capture efficiency and behavior of our sorbent. The main addition to the initial 298K baseline simulation code was the solve_system() function, which served as a method to couple up the two main processes so they could be solved simultaneously. These two processes being (1) the axial change in gas-phase CO_2 concentration along the height of the reactor, and (2) the radial diffusion and reaction of CO_2 inside each droplet.

The solve_system() function creates a radial grid for the particle and implements the aforementioned governing equations for CO_2 transport in spherical coordinates. The boundary conditions, defined in an earlier section, and equations are then passed to solve_bvp(), a numerical solver that computes the steady-state profiles of both gas-phase CO_2 and sorbent concentration. This lets the model determine how much CO_2 is removed at each point along the reactor. Because the solve_system() function allows for input parameters such as droplet radius, reaction rate, diffusivity, and inlet sorbent concentration, it can be used to explore different operating simulations and analyze the behaviour of the changes in parameters.

After running the simulation at the baseline temperature of 298K, we decided to add complexities to the model, starting with studying the temperature dependence of CO_2 removal. In chemical kinetics, most reactions become faster at higher temperatures because a greater fraction of molecules has enough energy to overcome the activation energy barrier; this relationship is captured by the Arrhenius equation, $k(T) = A*\exp(-E_a/RT)$. Diffusivity and reaction rate were updated to become temperature-dependent variables using the Arrhenius equation allowing us to sweep across different operating conditions and quantify how temperature affects absorption efficiency. Instead of using the full form, the code implements an Arrhenius scaling around a reference temperature (298 K), allowing the rate constant at any temperature to be computed from the chosen base value using $k(T) = k_{rxn}\exp(-E_a/R(T-1/T_{operating})$. This avoids the need to estimate the pre-exponential factor while still letting the observation for correct temperature sensitivity.

The code was expanded to then include calculations that determine how much $Ca(OH)_2$ sorbent is required to meet SciTech's 8-hour CO_2 capture requirement. This section is separate from the BVP solver and uses simple stoichiometry, taking into account the total CO_2 emissions,

the 90% capture requirement, and the 1:1 reaction between CO_2 and $Ca(OH)_2$. By converting moles to mass and then to slurry volume, the code outputs a clear and actionable sorbent requirement that can be reported. Additionally, we explored CO_2 capture efficiency and sorbent utilization with varying parameters of void fraction, particle radius, and initial sorbent concentration.

Results

Base Case – Temperature (298K)

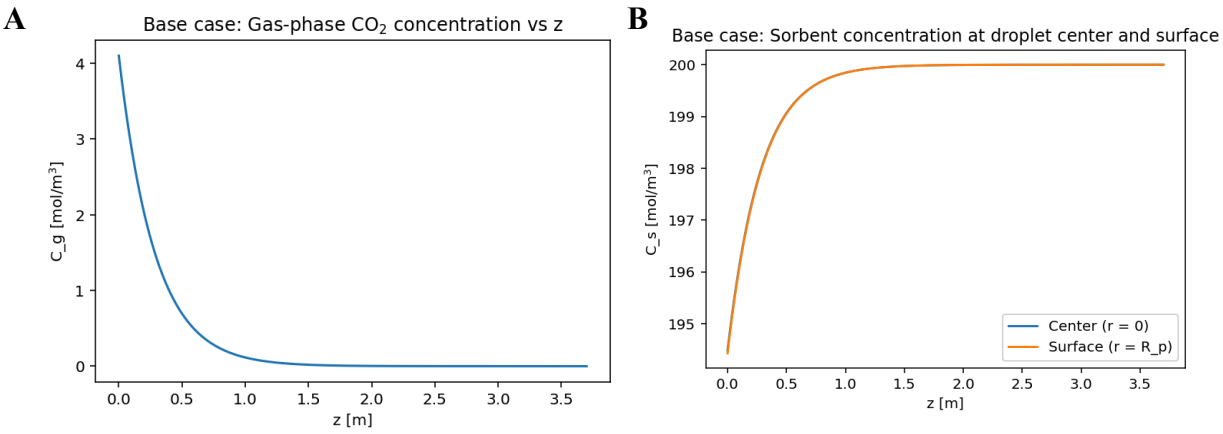


Figure 2. Base case simulation plot of Gas-Phase Concentration [mol/m^3] vs Height [m] (**A**) Base case simulation plot of Liquid-Phase Concentration [mol/m^3] vs Height [m] (**B**)

Under the base-case conditions, the model predicts a strong and rapid decrease in gas-phase CO_2 concentration along the reactor height (z). As shown in the first plot, the inlet CO_2 concentration begins at approximately 4.1 mol/m³ and drops sharply within the first meter of the reactor. By around z = 1.5m, the gas-phase concentration approaches nearly zero. This behavior shows that CO_2 is being absorbed efficiently into the $Ca(OH)_2$ slurry, consistent with the literature that describes carbonation of $Ca(OH)_2$ as fast and largely diffusion-controlled.

The second plot shows the sorbent ($Ca(OH)_2$) concentration inside each droplet at the center and at the surface. Both curves start slightly below the inlet value of 200 mol/m³ and increase with reactor height until they level out near 200 mol/m³. This indicates that only a very small amount of $Ca(OH)_2$ is consumed during absorption. The slight difference between center and surface concentrations near the reactor inlet reflects the expected diffusion gradient but the gradient disappears quickly as CO_2 used up in the gas phase.

It is also important to note that the blue line representing the droplet center is not visibly separate in the plot because it overlaps perfectly with the orange surface line. This happens because the reaction is extremely fast and CO_2 is removed very early in the reactor, leaving the droplet unreacted throughout most of its radial profile. As a result, the center and surface concentrations are nearly the same, causing the two curves to lie directly on top of each other.

Overall, the base-case results show that the reactor under these conditions is highly effective at CO_2 removal, with nearly complete gas-phase depletion well before the outlet. This behavior matches expectations for $Ca(OH)_2$ slurries and establishes a solid baseline for evaluating temperature effects and sorbent consumption in later simulations.

Complexity 1: Temperature dependence

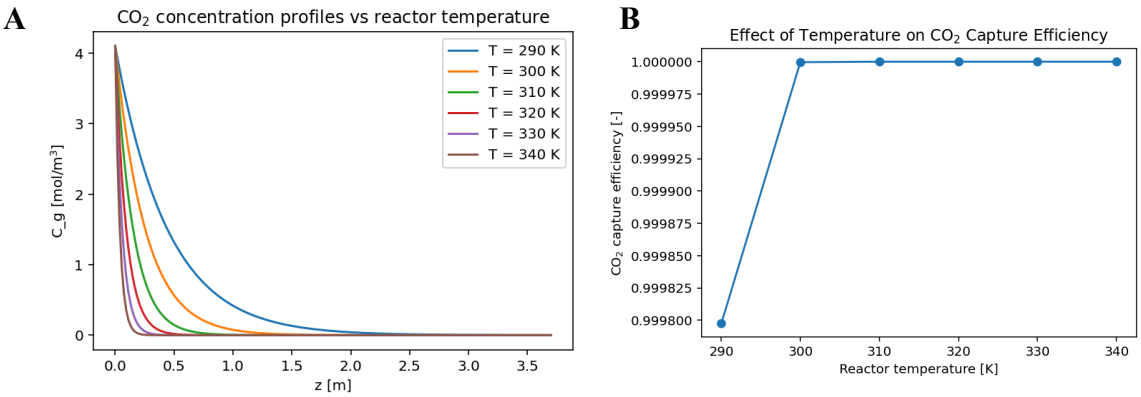


Figure 3. CO_2 Concentration Profiles vs Reactor Operating Temperature **(A)** CO_2 Capture Efficiency vs Reactor Operating Temperature **(B)**

For the interpretation of the temperature-dependent results, it is important to understand first the simplifying assumptions within the model, which result in the system having a nearly complete CO_2 removal. The carbonation reaction is modeled as extremely fast, the $Ca(OH)_2$ sorbent is present in large excess, mass-transfer resistances at the gas–liquid interface are neglected, and droplets are assumed to be small, spherical, and uniformly reactive. Together, these assumptions create an idealized gas-limited absorber where the rate of CO_2 removal is constrained almost entirely by how fast CO_2 reaches the droplets—not by how quickly the reaction occurs. Because the gas-phase inlet concentration is low relative to the sorbent

concentration, the model predicts very little sorbent depletion and nearly total CO_2 absorption along the reactor height.

The CO_2 capture efficiency used in Figure 4 reflects this ideal behavior. Capture efficiency was defined as

$$\eta = 1 - C_{g,out} / C_{g,in}$$

meaning it measures the fraction of incoming CO_2 removed by the time the gas exits the reactor. Since the model accounts for $C_{g,out}$ toward zero due to fast reaction and strong mass-transfer assumptions, the resulting efficiencies exceed 99.98% for all temperatures evaluated. At 290 K, the outlet concentration is already extremely small, corresponding to ~99.98% removal, and at temperatures above 300 K the model predicts complete depletion of CO_2 in the gas phase. These results indicate that the modeled system is already operating in a sense where temperature has little influence on the total fraction of CO_2 removed.

The CO_2 concentration profiles in Figure 3-A, give a more deep understanding of how temperature affects the shape of CO_2 removal. At lower temperature (290 K), the gas-phase CO_2 decays more gradually along the reactor, approaching zero near z = 1–1.5 m. As temperature increases, both diffusivity and reaction rate increase, causing CO_2 to be consumed much earlier in the reactor. For temperatures of 300–340 K, almost all CO_2 is absorbed within the first few tens of centimeters of reactor height. This demonstrates that temperature mainly shifts the location where absorption occurs, moving the reaction zone closer to the reactor inlet, but it does not meaningfully change the overall capture efficiency due to the idealized and sorbent-rich assumptions of the model.

<u>Complexity 2: Void Fraction</u>

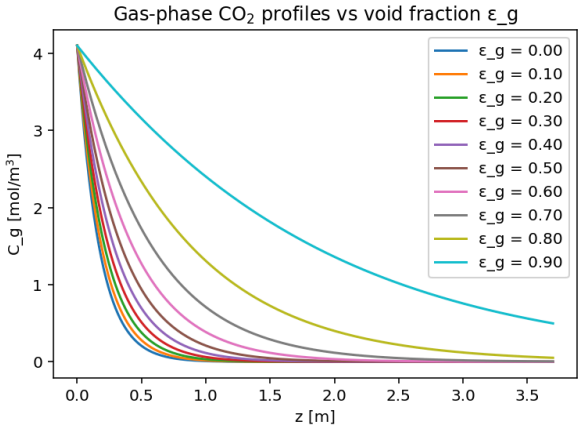
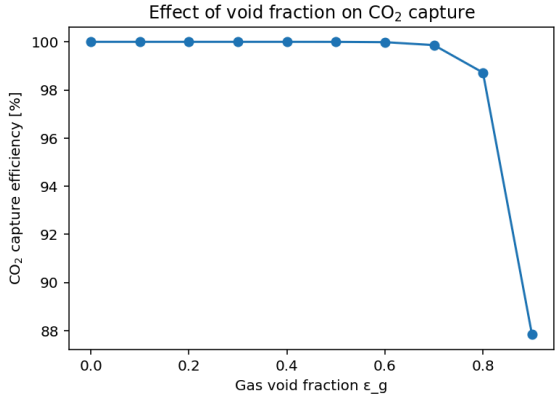


Figure 4. Gas-phase CO_2 profiles throughout reactor length with varying gas void fractions.

As mentioned, once we arrived at a working version of our model, we added complexities to it by varying different parameters. We first decided to explore the effects of increasing the gas fraction, which is analogous to decreasing the number of liquid sorbent droplets found within the reactor. We were curious about the optimization of this parameter as it could inform a minimum amount of sorbent to be used while still meeting the capture efficiency requirements, effectively allowing us to reduce the amount of sorbent waste produced. From Figure 5, we are able to visualize this effect and identify ε_g **= 0.80** as the most efficient gas fraction value tested. Any lower, and CO_2 is absorbed very early on in the reactor, indicating unreacted sorbent is present throughout most of the reactor. Any higher and C_g never fully reaches or approaches 0, which is unacceptable given the condition that there must be a capture efficiency of more than 90%. This is further explored in the following figure.

A



B

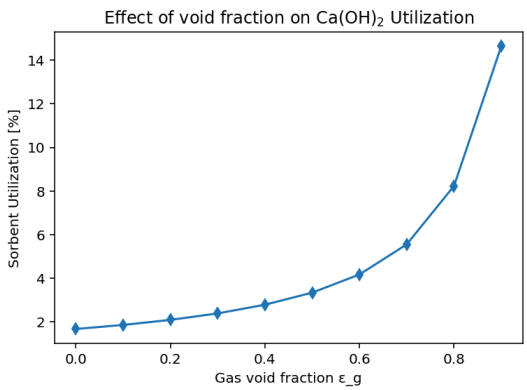


Figure 5. Effect of void fraction on gas capture efficiency (**A**) and liquid sorbent utilization (**B**).

The above figure confirms that for a desired capture efficiency above 90%, we must have a gas void fraction lower than 90%, so $\varepsilon_g = 0.80$ is a suitable choice. Connecting it back to the idea of reducing sorbent waste, Figure 6B. explores the relationship between sorbent utilization and gas void fraction. We see that increasing the void fraction increases the amount of sorbent used, as we predicted. Thus, by increasing the void fraction from the original 40% to 80% we see more than a 5% increase in sorbent utilization. These numbers are still relatively low, but we believe this is a good starting point to identifying which parameters could be further modified in future implementations of this model to make a fully functional, optimized capture reactor. An additional note is that these void fraction calculations are being performed with the underlying assumptions and parameters defined for the base; the complexities detailed in these results

section have not been compounded to compile a fully optimized complex case. It might be that with the right temperature and particle radius considerations, capture efficiency and sorbent utilization could be further optimized.

Complexity 3: Particle Radius

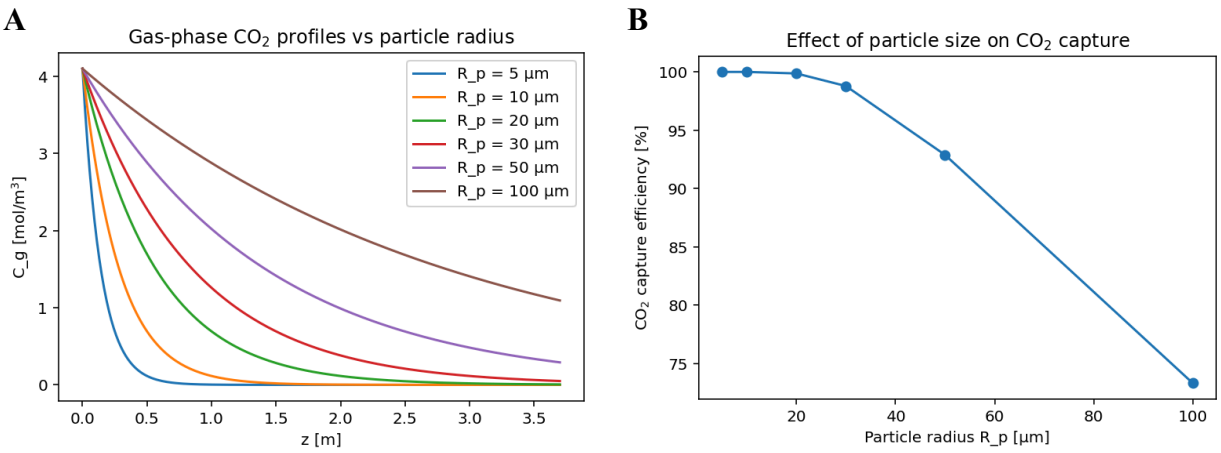


Figure 6. Gas-phase CO_2 profiles throughout reactor length with varying sorbent particle sizes

The last complexity explored in this report is in relation to the particle sizing and how it affects both the concentration profiles and the capture efficiency. The graphs presented in Figure 6 clearly show the effect of particle size on these two variables. As shown in Figure A, smaller particles (ranging from 5 - 20 µm) lead to a faster decrease of CO_2 concentration along the reactor's length. The gas is essentially captured within the first meter, a similar behaviour shown when varying the other parameters above. This indicates that the smaller droplets are very efficient in absorbing and reacting with the incoming CO_2 gas particles. This behavior can be explained as the smaller particles provide more surface area relative to their volume and allow CO_2 to be diffused through them more quickly.

As particle size increases, the behaviour starts to change noticeably. Bigger particles (ranging from 50 -100 µm) show that there is a much later depletion of CO_2 and even a significant amount of CO_2 remaining by the end of the reactor. This is due to the fact that the particle is so large that the CO_2 does not penetrate deeply enough to allow the sorbent to absorb it completely before the gas continues traveling upward.

This trend is demonstrated in Figure B, as the carbon capture efficiency decreases with increasing particle size. Keeping this in mind and thinking about the motivation for the project in relation to sustainability (minimizing sorbent waste), the optimized particle size would be **30 µm**. This would maintain a capture efficiency higher than 90% and the carbon concentration drops later, which results in essentially less unused sorbent.

A Note on Sorbent Utilization

The sorbent sizing calculation connects the model to the design requirement of capturing at least 90% of SciTech's CO_2 emissions over an 8-hour period. Using the boiler emission rate of 1273.37 mol CO_2/h, the total CO_2 produced in 8 hours is about 10,187 mol. To have 90% captured CO_2, the process must remove about 9,168 mol of CO_2. Because the carbonation reaction is 1:1 in moles between CO_2 and $Ca(OH)_2$, the same number of moles of $Ca(OH)_2$ is required, which corresponds to about **679 kg** of **pure $Ca(OH)_2$**. If we assume a slurry concentration of 200 mol/m³ (the same value used as the inlet sorbent concentration in the model), this amount of sorbent would be contained in approximately 45.8 m³ of slurry. This volume should be interpreted as the total amount of reactive slurry needed over 8 hours).

Conclusion and Future Directions

In conclusion, our team has identified the SciTech natural gas boiler as a major point source of CO_2 emissions on the Tufts University campus and proposed a calcium hydroxide slurry-based capture system as a solution to reduce these emissions. We established the basis for this approach through a review of the carbonation reaction, the limitations imposed by equilibrium and mass-transfer processes, and an analysis of SciTech's combustion-related CO_2 output using the building energy consumption data. We defined the governing equations that describe the CO_2 absorption and sorbent conversion and established the simplifying assumptions needed to create this system and for it to be applicable to our model conditions. We used the discretized forms of these equations alongside key parameters carefully selected via literature review to implement the PDE models in Python with a starter code provided by the instructor. The numerical analyses yielded valuable visualizations of the liquid and gas-phase concentration profiles along the reactor, and growing complexities were added to track the changes in said concentrations with changing temperature, void fraction, and droplet radius. We identified that

optimal reactor conditions would be at 290K, $\varepsilon_g = 0.80$, and $R_p = 30$ μm. However, these parameters were not tested in conjunction with each other, but rather in comparison to our original base case. Further implementations and improvements of this work should make an attempt at putting all of these optimized parameters together and obtain a working model that both maintains efficient CO_2 removal and produces less sorbent waste. Additionally, further complexities could be added to this model by considering potential sorbent recycling as part of the reactor design. This would challenge the current steady state assumptions built-in to the reactor model, but would likely be helpful in reducing the initial amount of slurry into the reactor and could improve efficiency over time. We could also consider modifying the defined contact time that the gas spends in the reactor and study capture efficiency and sorbent utilization variance with this variable.

We have identified these areas for growth for our working code, but we believe that we have already built a strong platform to understand what could be done to reduce SciTech's carbon footprint and work towards a more sustainable future. Using $Ca(OH)_2$ as a capture sorbent, we have demonstrated a potentially efficient and effective way of tackling this problem and offered future solutions to make it more applicable with real-world constraints.

APPENDIX

Critical Parameter Selection - Expanded

Gas flow and volumetric flow rate

This parameter was defined directly from the SciTech boiler CO_2 generation rate as detailed in an earlier section. As established previously, the boiler produces 1273.37 mol CO_2/hr, and assuming the flue gas contains 10 mol% CO_2 gives a total flue gas flow of 12733.7 mol/hr. After converting this to mol/s resulted in n = 5.54 mol/s. Assuming ideal gas for the determination of the volumetric flow rate and at 298 K and 1 atm, this leads us to a volumetric flow of $V = 0.0865$ m^3/s.

Reactor size (Diameter, Length and Volume)

The reactor size was chosen by using the same approach as Kakaraniya et al. in "*Reactive Precipitation in Gas-Slurry Systems: The CO_2−$Ca(OH)_2$−$CaCO_3$ System*." We used their proposed values and performed calculations to create a scaling factor to determine an appropriate reactor size for our conditions. Their reactor had D = 0.09 m and H = 0.26 m, which corresponds to a volume of 1.65×10^{-3} m^3. With a gas flow rate of 1.33×10^{-4} m^3/s and a $\varepsilon_g \approx 0.40$, the gas residence time is about 4.9 s. We preserved that same gas residence time and void fraction when scaling to the SciTech boiler conditions by computing the required gas volume $V_{gas,SciTech}=V_{gas}t_{contact}$ and then total reactor volume $V_{reactor}=V_{gas}/\varepsilon_g = 1.06$ m^3. Choosing a diameter of 0.60 m then sets the height to L=3.7 m. Using this approach, our reactor is large enough to be installed in SciTech but dynamically similar in conditions to the system studied in the literature.

Sorbent particle size, (R_p = radius of droplet)

Kakaraniya et al. reports $Ca(OH)_2$ particle sizes in the micrometer range and explore different radii up to 10 μm for their model. We chose R_p = 10 μm as our droplet radius because it lies within the studied range and still represents a "fine" particle.

Operating Temperature

We set the base operating temperature to 298 K (25 °C) because both main reference studies that applied to our model, Kakaraniya et al. and Han et al, perform their experiments and modeling at this temperature. The decision to use 298 K as the reference also simplifies the parameter selection: the reported diffusivity, solubility, and reaction behavior are all given at or near 25 °C,

so we can use their values directly without additional temperature variations. The underlying simplifying assumption is that we can treat temperature as a constant value and neglect variance along the reactor; we might assume that there is access to temperature controlling units like heat exchangers or coolers to maintain these conditions.

Gas void fraction ε_g

As mentioned previously, the gas void fraction $\varepsilon_g = 0.40$ was chosen by matching the gas void fraction from the Kakaraniya stirred contactor. For their geometry and gas flow, taking ε_g to be 0.40 results in a gas contact time of 4.9 seconds, which we used as our target contact time when scaling up to the SciTech reactor. Using $\varepsilon_g = 0.40$ in our model means that 40% of the reactor volume is gas and 60% is occupied by slurry droplets, which is reasonable for the gas–liquid contact and makes sure that our scaled reactor has similar behavior to the reference system.

Diffusivity of CO_2 in the sorbent phase D

We used $D = 2.0\times10{-}9$ m^2/s as the diffusivity of CO_2 in the liquid/droplet phase directly from the property table used by Kakaraniya et al., who in addition cites Jones et al. (1992).

Reaction rate constant k_{rxn}

The literature describes the CO_2–$Ca(OH)_2$ reaction in aqueous media as very fast and often treats it as instantaneous at the solid–liquid interface, with overall rates controlled by mass transfer and diffusion. In Kakaraniya's shrinking-core formulation, the reaction at the interface is almost infinitely fast and no explicit k_{rxn} appears; instead, the model is driven by diffusion and external mass transfer coefficients. We chose an initial value $k_{rxn} = 3\times10^{-8}$ $m^5/(mol^2 \cdot s)$, which is large enough that the Damköhler $\gg 1$, meaning the system remains effectively diffusion-controlled and consistent with the "very fast reaction" description.

Gas velocity

The gas velocity was calculated directly from the SciTech boiler's volumetric flue-gas flow rate. Using the ideal-gas–based volumetric flow of $V_{gas} = 0.0865$ m^3/s and dividing by the reactor cross-sectional area A = 0.283, we obtained a gas velocity of 0.306 m/s. We rounded this value to $u_g = 0.3$ m/s.

Liquid Velocity

The liquid velocity magnitude was chosen to be similar to the gas velocity ($|u_l| = 0.3$ m/s) in order to give both phases comparable residence times through the reactor. The negative sign indicates counter-current operation, where the $Ca(OH)_2$ slurry flows downward while the gas flows upward.

Fresh sorbent concentration

We chose a fresh sorbent concentration of $C_{s0} = 200$ mol/m^3, which corresponds to approximately 0.20 mol/L (15 g/L) of $Ca(OH)_2$. This value is representative of a moderately concentrated $Ca(OH)_2$ slurry, typical of industrial absorption applications. It is large enough to make sure that sorbent availability is not immediately limiting while still remaining physically realistic for an aqueous slurry.

Inlet CO_2 concentration

The inlet gas-phase CO_2 concentration was calculated from the assumed flue-gas composition of 10 mol% CO_2 at 1 atm and 298 K. Using the ideal-gas relation $C = p/(RT)$, the partial pressure of CO_2 (10132.5 Pa) gives a concentration of 4.09 mol/m^3, which we rounded to $C_{g,in} = 4.1$ mol/m^3.

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